

# Gingerbread Man / Guitar Hero Controller

Hector Lugaro

# What?

- Gingerbread man that screams when its limbs are torn off
- USB C 2.0 Guitar Hero Controller, with battery charging



# Why?

- Kind of funny, gag gift perhaps?
- Guitar Hero is awesome, but controllers online are kinda expensive, limited PC options with wireless. I've had this in the back of my mind for years.

Guitar Hero Controller Pc 531+ items with ads

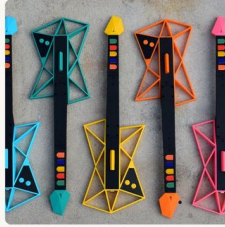
Exclude digital items Personalizable Under \$50 Free shipping Arrives within 7 days Ships from United States Get by Mother's Day Sort by: Relevancy 11



Clone Hero BetterGH3 Guitar Controller  
★★★★★ (177)  
\$84.99 \$99.99 (15% off)



DIY Modding Kit for Guitar Hero Controllers (RGB)  
★★★★★ (374)  
\$29.00  
RGB Fretboard Available!



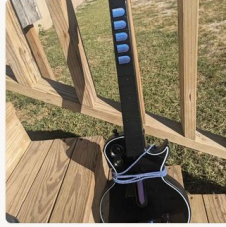
Custom Clone Hero Controller  
★★★★★ (319)  
\$150.00



Clone Hero Mech modded Guitar Controller Black...  
★★★★★ (14)  
\$183.75



Whammy Re-Design UniStrum Custom Guitar Con...  
★★★★★ (108)  
\$100.00  
THE 1 FOOT LONG GUITAR CONTROLLER  
MAKE IT YOURS.  
PC CONSOLE MOBILE  
BUNDLE INCLUDES  
THE UNISTRUM, A HEADSTOCK,  
& A BODY ATTACHMENT!



Modded Guitar Hero/Clone Hero Les Paul Mechan...  
★★★★★ (167)  
\$140.00  
FREE shipping



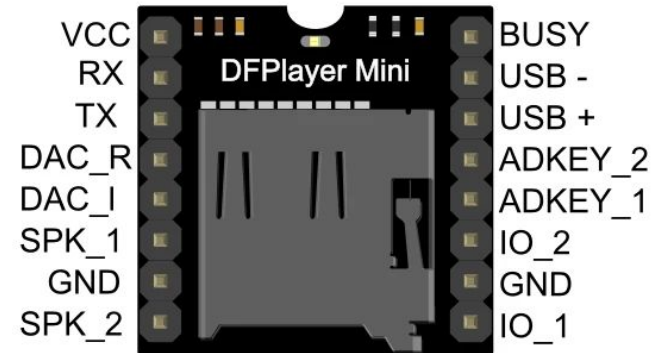
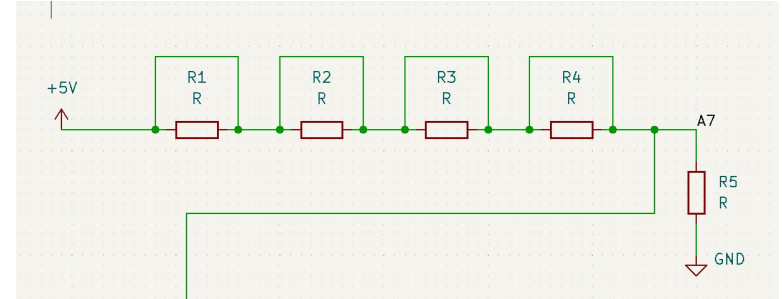
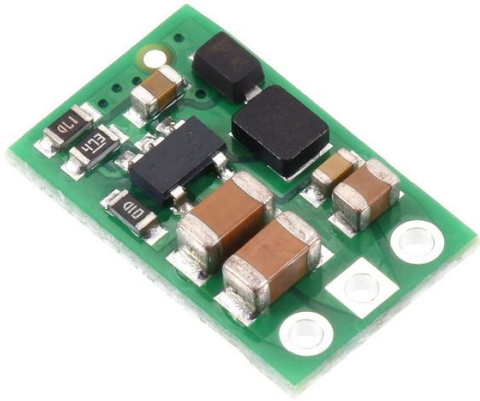
Modded Guitar Hero/Clone Hero Controller: Mech...  
\$140.00



Strum Bar Holder For Guitar Hero Guitars LP, SG, X...  
★★★★★ (184)  
\$4.99

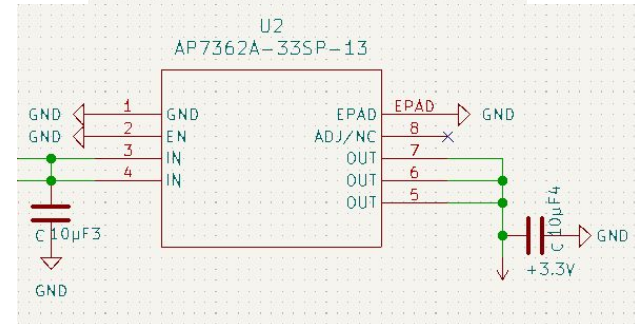
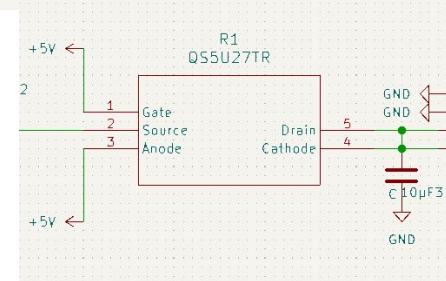
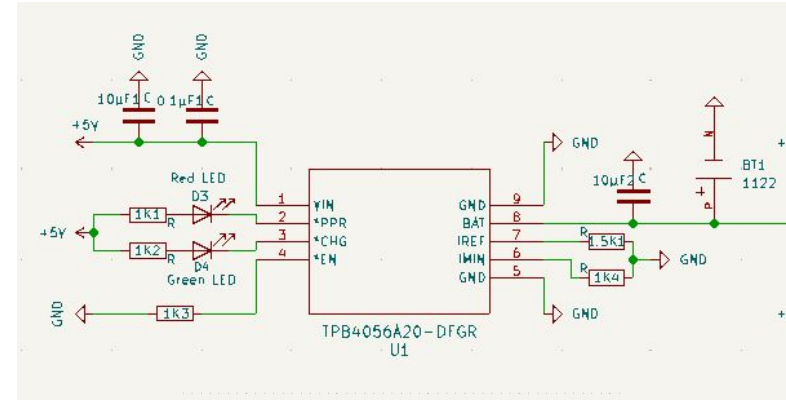
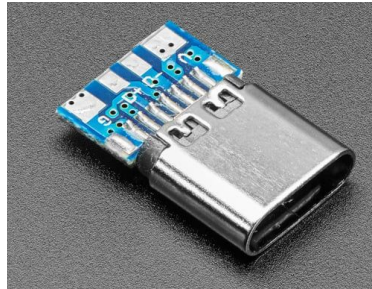
# How?

- 5 Resistors in parallel with wires.
- MP3 Player Board
- 5V Step Up Module



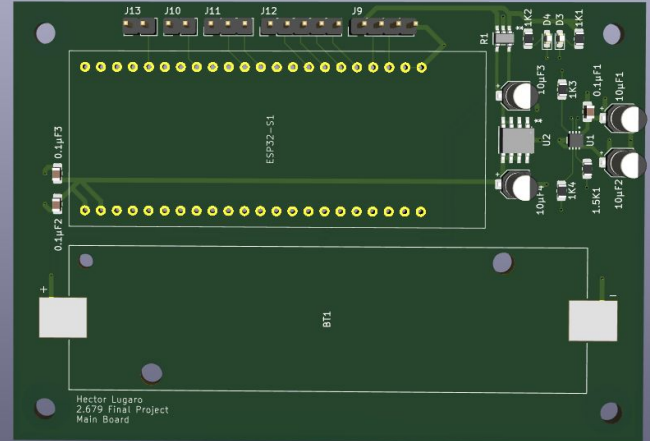
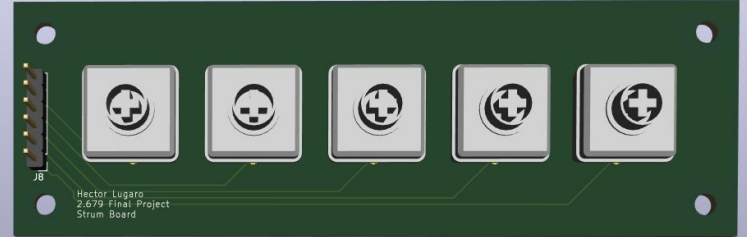
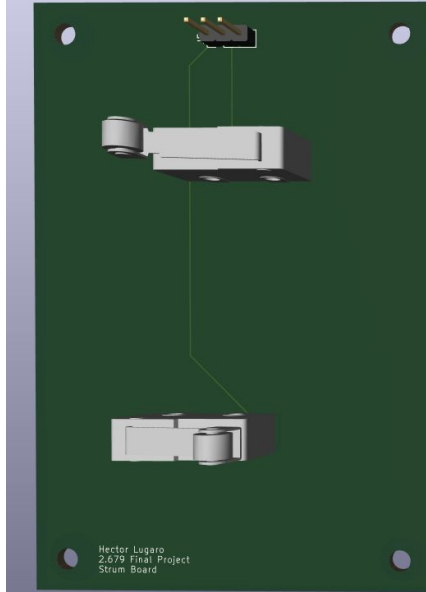
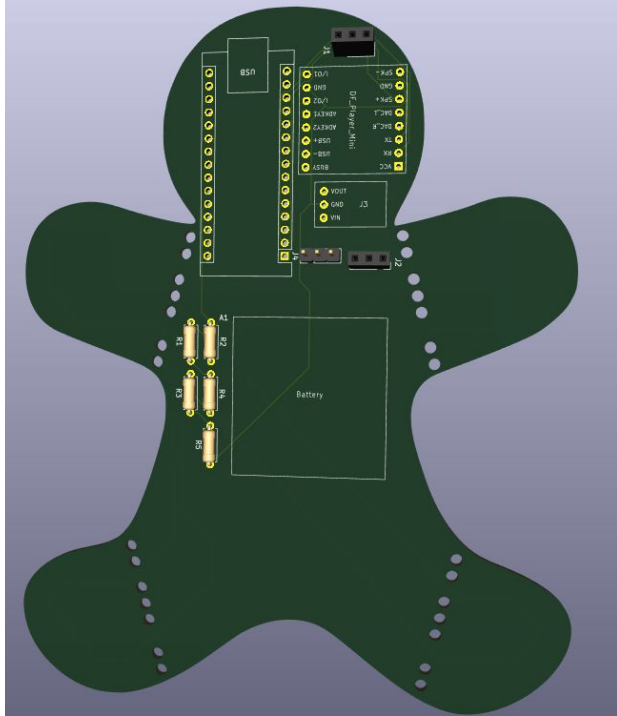
# How? pt2

- 3 Different Boards: Strum, Frets, Main
- ESP-32 S3 C1 can be Human Input Device, can directly read USB data
- 3 Different ICs: TP4056 (Battery Charger), P-Channel MOSFET w. Shottky Diode, LDO
- USB-C Socket Breakout



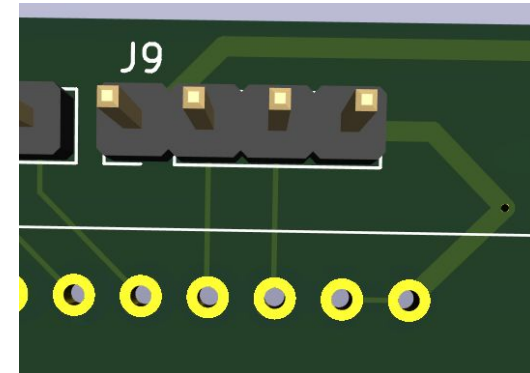
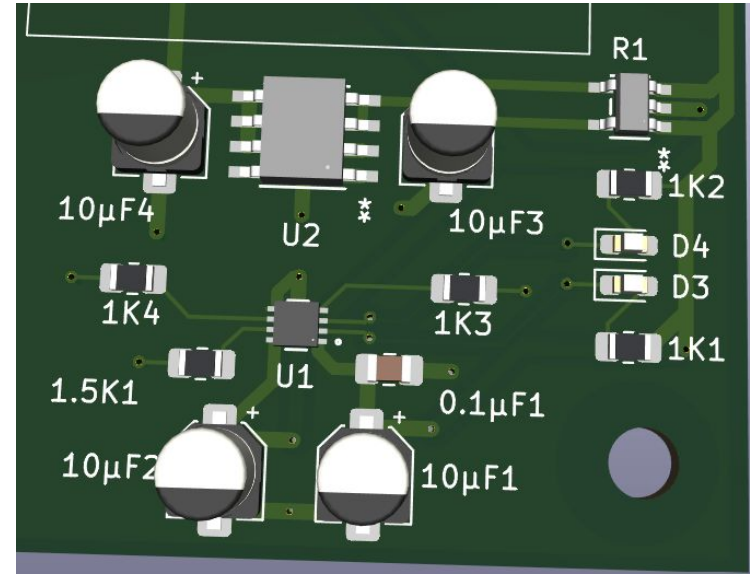


# Pics!



# Special Stuff!!!

- Signal noise!
- Bypass and Electrolytic Capacitors!
- Thicker Traces
- Keep out zones
- Battery Protection
- Vias!



# Testing

- Check that audio playback works as intended on Gingerbread Man
- Test Controller on just battery, just USB, and both at the same time.
- Convert button presses, strums, into HID inputs.
- How make guitar?